

Spleen and Lymphatic System

Spleen and clonal haematopoiesis (JAK2), lymphatic vessels (LYVE1), lymphocyte count and predisposition to lymphomas (HLA, IRF4, REL)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Non-Hodgkin lymphomas have a very low genetic predisposition: the consensus of the scores with good coverage is at percentile **13.8**, the non-Hodgkin lymphoma score at percentile 1.1, the follicular lymphoma score at percentile 0.2, and the burden of 47 loci at **0.0**. The two HLA class II loci for follicular lymphoma carry no risk alleles (C/C, T/T). Hodgkin lymphoma is also unlikely (consensus at percentile 32.3, loci burden at percentile 18.0). In the spleen and bone marrow, the **JAK2 46/1 haplotype** is absent (C/C): no inherited predisposition to the acquired JAK2 V617F mutation, the most common cause of an enlarged spleen due to a myeloproliferative neoplasm and one form of clonal haematopoiesis. The loci burden for spleen diseases is at percentile 18.7.

Unfavourable / to watch. Chronic lymphocytic leukaemia has its score at percentile 78.4 (typical, on the high side), with 2 alleles in **ACOXL/BIM** (G/G) and 1 in IRF4; the burden of its 9 loci is at percentile 28.0. For Hodgkin lymphoma there is 1 allele each in HLA class II, REL and PVT1, and 2 in GATA3, with no overall effect (low consensus).

No judgement (descriptive). The lymphocyte count has typical-to-high genetics (consensus at percentile 75.4). The lymphatic vessels (levels of LYVE1, the hyaluronic acid receptor of lymphatic vessels) have a loci burden at percentile 21.0.

What it means. The lymphatic system and the spleen have a favourable genetic profile: lymphomas are unlikely (non-Hodgkin very unlikely) and there is no JAK2 46/1 predisposition. This matches the Cancers One by One report. There are no genetic scores for spleen size or for common lymphoedema; the spleen reading rests on JAK2 and on the spleen disease loci.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality DR2 ≥ 0.8 . Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 14 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 11 from the chip, 3 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 2 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 2,867,733 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 203 loci, 132 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Non-Hodgkin lymphomas	13.8	4 of 5	—	0
Hodgkin Lymphoma	32.3	2 of 3	39.3	1
Lymphocytes	75.4	3 of 3	51.9	2

For lymphomas and leukaemia, a higher percentile means more risk. For the lymphocyte count, the percentile is descriptive only. The lymphoma scores have few variants (5 to 43), except the genomic Hodgkin score; two of them (64% and 68% coverage) are left out of the consensus.

Non-Hodgkin lymphomas

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Non-Hodgkin lymphoma	PGS004248 (Kim 2023, 20 variants)	15 of 20 (coverage 64%)	26.2	TYPICAL	typical
Non-Hodgkin lymphoma	PGS000080 (Graff 2020, 19 variants)	18 of 19 (coverage 96%)	1.1	LESS LIKELY	favourable
Diffuse large B-cell lymphoma	PGS003454 (Berndt 2022, 5 variants)	4 of 5 (coverage 84%)	26.5	TYPICAL	typical
Follicular lymphoma	PGS003455 (Berndt 2022, 7 variants)	7 of 7 (coverage 100%)	0.2	LESS LIKELY	favourable
Chronic lymphocytic leukaemia	PGS003453 (Berndt 2022, 43 variants)	38 of 43 (coverage 87%)	78.4	TYPICAL	typical

Hodgkin lymphoma

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hodgkin lymphoma (genomic)	PGS000637 (Fritsche 2020, 1.05 million variants)	1,020,080 of 1,047,511 (coverage 97%)	39.3	TYPICAL	typical
Hodgkin lymphoma	PGS003449 (PGS Catalog 2023, 21 variants)	16 of 21 (coverage 68%)	0.8	LESS LIKELY	favourable
Hodgkin lymphoma	PGS000639 (Fritsche 2020, 20 variants)	18 of 20 (coverage 89%)	25.3	TYPICAL	typical

Lymphocytes

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lymphocyte count (genomic)	PGS004356 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	28.4	TYPICAL	no judgement
Lymphocyte count (genomic)	PGS002183 (Privé 2022, 815,000 variants)	799,750 of 814,921 (coverage 97%)	75.4	TYPICAL	no judgement
Lymphocyte count	PGS000097 (Xu 2022, 25,000 variants)	22,148 of 24,646 (coverage 91%)	81.0	HIGH	no judgement

4. Key variants by topic

Variants for the predisposition to clonal haematopoiesis with an enlarged spleen (JAK2 46/1), for LYVE1 levels in lymphatic vessels, and replicated loci for Hodgkin lymphoma, non-Hodgkin lymphomas and chronic lymphocytic leukaemia.

Spleen, bone marrow and clonal haematopoiesis

The JAK2 46/1 haplotype makes the acquired V617F mutation in blood stem cells more likely.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
JAK2	rs10974944	46/1	C/C	imputed DR2=1.00	G (0 copies)	favourable	JAK2 46/1 haplotype: the G allele predisposes to the acquired JAK2 V617F mutation (myeloproliferative neoplasms with enlarged spleen and clonal haematopoiesis); ~3× per copy	Strong

Lymphatic vessels

Descriptive genetics of LYVE1, a marker of lymphatic vessels (no judgement).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LYVE1/ IRAG1	rs35468145		G/G	chip	A (0 copies)	no judgement	LYVE1 region (hyaluronic acid receptor of lymphatic vessels): levels of LYVE1 [GWAS Catalog: Lymphatic vessel endothelial hyaluronic acid receptor 1 levels, allele A, beta 0.417 [0.35-0.48] unit increase, p=2E-38, PMID 34648354]	Weak
B4GALT1	rs7865362		C/C	chip	T (0 copies)	no judgement	B4GALT1 (glycosylation): levels of LYVE1 [GWAS Catalog: LYVE1 protein levels, allele T, beta 0.107 [0.095-0.119] unit increase, p=9E-79, PMID 39789286]	Weak

Hodgkin lymphoma

HLA class II, NF-κB (REL) and T-cell differentiation (GATA3).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA- DRB9 (class II)	rs6903608		C/T	chip	G (1 copy)	unfavourable (mild)	HLA class II region: the largest locus of Hodgkin lymphoma [GWAS Catalog: Hodgkin's lymphoma, allele G, OR 1.70, p=3E-50, PMID 21037568]	Strong
REL	rs1432295		G/A	chip	G (1 copy)	unfavourable (mild)	REL (NF-κB): Hodgkin lymphoma [GWAS Catalog: Hodgkin's lymphoma, allele G, OR 1.22, p=2E-8, PMID 21037568]	Moderate
PVT1	rs2019960		T/C	chip	G (1 copy)	unfavourable (mild)	PVT1/MYC (8q24): Hodgkin lymphoma [GWAS Catalog: Hodgkin's lymphoma, allele G, OR 1.33, p=1E-13, PMID 21037568]	Moderate
GATA3	rs3781093		T/T	imputed DR2=0.97	T (2 copies)	unfavourable	GATA3 (T lymphocyte differentiation): Hodgkin lymphoma [GWAS Catalog: Nodular sclerosis Hodgkin lymphoma, allele T, OR 1.39, p=9E-13, PMID 29196614]	Moderate

Non-Hodgkin lymphomas and chronic lymphocytic leukaemia

HLA class II (follicular lymphoma), IRF4 and B-cell apoptosis (BIM).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA-DRB5	rs12195582		C/C	chip	T (0 copies)	favourable	HLA class II region: follicular lymphoma (~1.8× per copy) [GWAS Catalog: Follicular lymphoma, allele T, OR 1.78, p=5E-100, PMID 25279986]	Strong
HLA-DQB1	rs10484561		T/T	chip	G (0 copies)	favourable	HLA-DQB1: follicular lymphoma (~1.9×) [GWAS Catalog: Follicular lymphoma, allele G, OR 1.95, p=1E-29, PMID 20639881]	Strong
EXOC2/ IRF4	rs116446171		C/C	imputed DR2=0.99	G (0 copies)	favourable	EXOC2/IRF4: a rare variant of large effect in lymphoplasmacytic lymphoma (Waldenström) [GWAS Catalog: Waldenström macroglobulinemia / lymphoplasmacytic lymphoma, allele G, OR 21.14, p=1E-54, PMID 30305637]	Moderate
IRF4	rs872071		A/G	chip	G (1 copy)	unfavourable (mild)	IRF4 3'UTR: chronic lymphocytic leukaemia [GWAS Catalog: Chronic lymphocytic leukemia, allele G, OR 1.54, p=2E-20, PMID 18758461]	Strong
ACOXL/ BCL2L11	rs17483466		G/G	chip	G (2 copies)	unfavourable	ACOXL/BIM (apoptosis): chronic lymphocytic leukaemia [GWAS Catalog: Chronic lymphocytic leukemia, allele G, OR 1.37, p=4E-17, PMID 23770605]	Moderate
SP140	rs13397985		T/T	chip	G (0 copies)	favourable	SP140 (nuclear lymphocyte protein): chronic lymphocytic leukaemia [GWAS Catalog: Chronic lymphocytic leukemia, allele G, OR 1.45, p=1E-22, PMID 23770605]	Moderate
GRAMD1B	rs735665		G/G	chip	A (0 copies)	favourable	GRAMD1B: chronic lymphocytic leukaemia [GWAS Catalog: Chronic lymphocytic leukemia, allele A, OR 1.62, p=4E-39, PMID 23770605]	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for lymphomas, chronic lymphocytic leukaemia, spleen diseases and LYVE1 levels. Alleles are oriented towards more risk (or towards higher levels, for LYVE1) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Hodgkin lymphoma	81 (39)	33 of 78	39	18.0	favourable
Non-Hodgkin lymphomas	54 (47)	24 of 94	47	0.0	favourable
Chronic lymphocytic leukaemia	9 (9)	7 of 18	9	28.0	typical
Spleen (splenomegaly and spleen diseases)	29 (10)	6 of 20	9	18.7	favourable
Lymphatic vessels (LYVE1)	30 (27)	19 of 54	27	21.0	no judgement

6. Favourable

- **Non-Hodgkin lymphomas:** consensus at percentile **13.8**; burden of 47 loci at **0.0**; follicular lymphoma at percentile 0.2; HLA class II and EXOC2/IRF4 without risk alleles. **Moderate to strong.**
- **Hodgkin lymphoma:** consensus at percentile 32.3; loci burden at percentile 18.0. **Moderate.**
- **JAK2 46/1 absent** (C/C): less predisposition to the JAK2 V617F mutation (enlarged spleen, polycythaemia, thrombocythaemia). **Strong.**
- **Spleen diseases:** loci burden at percentile 18.7 (few loci read). **Weak.**
- **SP140** and **GRAMD1B** (chronic lymphocytic leukaemia) without risk alleles. **Moderate.**

7. Unfavourable

- **Chronic lymphocytic leukaemia:** score at percentile **78.4** (typical, high side); ACOXL/BIM G/G (2 alleles) and IRF4 A/G (1 allele). **Weak.**
- **Hodgkin, isolated alleles:** GATA3 T/T (2 alleles), and HLA class II, REL and PVT1 with 1 allele; no overall effect. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Clonal Haematopoiesis (CHIP)	JAK2 46/1 absent: no inherited predisposition to JAK2 V617F clonal expansion	Favourable
Immunosenescence	B-cell immune surveillance: non-Hodgkin lymphomas with consensus at percentile 13.8 and loci at percentile 0.0	Favourable
Somatic Mutation Accumulation	Lymphomas and chronic lymphocytic leukaemia depend on somatic mutations in lymphocytes; low inherited predisposition (except a typical CLL score)	Favourable
Loss of Naive T Cells	Lymphocyte count with typical-to-high genetics (consensus at percentile 75.4)	Typical
Chronic Inflammation	Lymph nodes and the spleen filter systemic inflammation; no genetic signs of splenomegaly	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Full blood count with differential** in routine blood tests: it is the simple screen for chronic lymphocytic leukaemia (persistent lymphocytosis) and myeloproliferative neoplasms. *Suggestion (not applied).*
2. **Signs not to ignore:** enlarged, painless lymph nodes for more than 4 weeks, night sweats, unexplained fever, unexplained loss of body mass, or discomfort or fullness on the left side of the abdomen. *Suggestion (not applied).*
3. **Vaccines** (pneumococcal, flu, shingles) kept up to date: the spleen and lymph nodes are the centre of the vaccine response. *Suggestion (not applied).*
4. **Avoid benzene and solvents** without protection (exposure linked to leukaemias and lymphomas). *Suggestion (not applied).*
5. **Already in the plan and protective:** daily exercise (improves lymphatic drainage), sleep and glucose control. *Suggestion (not applied).*

10. Limitations

- The lymphoma scores have few variants; the agreement between scores, consensus and loci burden gives more confidence.
- There are no published genetic scores for spleen size or common lymphoedema; hereditary lymphoedema depends on rare mutations (FLT4, FOXC2, GJC2, PIEZO1) that the chip does not read.
- The JAK2 V617F mutation is acquired; the 46/1 haplotype only measures the inherited predisposition.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (11 scores: non-Hodgkin lymphomas, Hodgkin lymphoma, chronic lymphocytic leukaemia and lymphocyte count). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 5 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Kilpivaara 2009, Olcaydu 2009 and Jones 2009 (JAK2 46/1), Sud 2017 (Hodgkin lymphoma), Skibola 2014 (follicular lymphoma), Berndt 2016 (chronic lymphocytic leukaemia), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Spleen, bone marrow and clonal haematopoiesis	JAK2	rs10974944	C/C	imputed DR2=1.00	G	0	+ strand (palindromic; + assumed)	without the risk allele
Lymphatic vessels	LYVE1/ IRAG1	rs35468145	G/G	chip	A	0	+ strand	0 copies of the effect allele
Lymphatic vessels	B4GALT1	rs7865362	C/C	chip	T	0	+ strand	0 copies of the effect allele
Hodgkin lymphoma	HLA-DRB9 (class II)	rs6903608	C/T	chip	G	1	complement (– strand)	1 copy of the risk allele
Hodgkin lymphoma	REL	rs1432295	G/A	chip	G	1	+ strand	1 copy of the risk allele
Hodgkin lymphoma	PVT1	rs2019960	T/C	chip	G	1	complement (– strand)	1 copy of the risk allele
Hodgkin lymphoma	GATA3	rs3781093	T/T	imputed DR2=0.97	T	2	+ strand	2 copies of the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	HLA-DRB5	rs12195582	C/C	chip	T	0	+ strand	without the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	HLA-DQB1	rs10484561	T/T	chip	G	0	+ strand	without the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	EXOC2/ IRF4	rs116446171	C/C	imputed DR2=0.99	G	0	+ strand (palindromic; + assumed)	without the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	IRF4	rs872071	A/G	chip	G	1	+ strand	1 copy of the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	ACOXL/ BCL2L11	rs17483466	G/G	chip	G	2	+ strand	2 copies of the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	SP140	rs13397985	T/T	chip	G	0	+ strand	without the risk allele
Non-Hodgkin lymphomas and chronic lymphocytic leukaemia	GRAMD1B	rs735665	G/G	chip	A	0	+ strand	without the risk allele